EtherCAT SubDevice Controller (ESC) Overview

as of July 2025

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Name	M76P00	M73P00	AX58100	AX58200	AX58400	ET1100	ET1810/ET1811/ET1812	ET1815/ET1816/ET1817	ET1820/ET1821/ET1822
Type	Dual-Core RISC-V SOC	Single-Core RISC-V SOC	ASIC	ARM MPU	Dual-Core ARM MPU	ASIC	Altera FPGA + IP Core	AMD (Xilinx) FPGA + IP Core	Microchip FPGA + IP Core
Supplier	匠芯创科技 ArtInChip	匠芯创科技 ArtInChip	ASIX	ASIX	ASIX	BECKHOFF	BECKHOFF	BECKHOFF	BECKHOFF
Supplier	Artinchip Guangdong Jiangxinchuang Technology	Artinchip Guangdong Jiangxinchuang Technology	ASIX Electronics Corporation	ASIX Electronics Corporation	ASIX Electronics Corporation	Beckhoff Automation	Beckhoff Automation	Beckhoff Automation	Beckhoff Automation
Package	BGA196, 0.8 mm pitch eLQFP144, 0.5 mm pitch BGA144, 0.8 mm pitch	eLQFP144, 0.5 mm pitch BGA144, 0.8 mm pitch	80-pin LQFP 0.4 mm pitch	144-pin HSFBGA 0.8 mm pitch	225LD EHS-TFBGA 0.8 mm pitch	BGA128 0.8 mm pitch	FPGA dependent	FPGA dependent	FPGA dependent
Size	BGA196: 12X12 mm eLQFP144: 20x20 mm BGA144: 10x10 mm	eLQFP144,: 20x20 mm BGA144: 10x10 mm	12 x 12 mm	10 x 10 mm	13 x 13 mm	10 x 10 mm	FPGA dependent	FPGA dependent	FPGA dependent
µC Interface	µC bus (internal, AHB)	µC bus (internal, AHB)	SPI/parallel (8/16-bit, asynchronous)	uC bus (Internal, AHB)	uC bus (Internal, AHB)	serial/parallel (8/16bit, sync/async)*	serial/parallel (8- /16-/32-bit, async) AVALON®* or AMBA-AXI4-Interface	serial/parallel (8- /16-/32-bit, async) OPB®* and PLB®*or AMBA-AXI4-Interface	serial/parallel (8- /16-/32-bit, async) AMBA-AXI4-Interface
Digital I/O	-	-	32	-	20	8-64*	8-64*	8-64*	8-64*
General Purpose I/O	-	-	32	up to 76*	up to 97*	0-32*	0-128*	0-128*	0-128*
DPRAM	8 Kbyte	8 Kbyte	9 kByte	9 kByte	9 kByte	8 kByte	0...60 kByte*	0...60 kByte*	0...60 kByte*
SyncManager Entities	8	8	8	8	8	8	0...16*	0...16*	0...16*
FMMU Entities	8	8	8	8	8	8	0...16*	0...16*	0...16*
Distributed Clock Support	yes(64 Bit)	yes(64 Bit)	yes (64-bit)	yes (64-bit)	yes (64-bit)	yes	yes*	yes*	yes*
No. of Ports	2-3 (RMII/MII)	2 (RMII/MII)	2 (100BaseTX) + Opt. 1 (MII)	2 (100BaseTX) + Opt. 1 (MII)	2 (100BaseTX) + Opt. 1 (MII)	2-4 (MII/E-BUS)*	1-4 (MII/RMII/RGMII)	1-4 (MII/RMII/RGMII)	1-4 (MII/RMII/RGMII)
Specials	Dual-Core 552MHz RISC-V with FPU and DSP,1 MB on-chip SRAM, USB FS OTG,additional RMII/MII Ethernet MAC with IEEE1588,1xCordic,1xFFT,2xHCL, 1xPFM,1xDCE,4xGPT,2x14bit 2Msps 24CH ADC,4xPGA,24CH EPWM/HRPWM,6xCAP,10xQEP, 2xQOUT,2xCPM,4xSDFM,1xPBUS,9xUART,3xQSPI,5xSPI,4xI2C,1xCAN,2xCANFD Temperature rated (-40 to +105°C)	Single-Core 552MHz RISC-V with FPU and DSP,512 KB on-chip SRAM, USB FS OTG,additional RMII/MII Ethernet MAC with IEEE1588,1xCordic,1xFFT,1xHCL, 1xPFM,1xDCE,4xGPT,2x14bit 2Msps 16CH ADC,4xPGA,24CH EPWM /HRPWM,6xCAP,10xQEP,2xQOUT,2xCPM,4xSDFM,1xPBUS,9xUART,2xQSPI,4xSPI,4xI2C,1xCAN,2xCANFD Temperature rated (-40 to +105°C)	100BASE-FX support 2 integrated PHYs 3-ch PWM and S/D I/F ABZ and Hall encoder I/F SPI master I/F	2 integrated Ethernet PHYs, USB 2.0 HS OTG, 10/100Mbps Ethernet MAC with RMII and hardware cryptography accelerator 6xLPUARTs, 3xISO-7816-3, 1xQuad-SPI, 3xI2C, 1xI2S, 2xUSCI, 2xCAN, 1xSPI Flash I/F, 2xSDHC, 1x16-ch/12-bit ADC, 2x12-bit DAC, 2xAnalog Comparators, 2xOperational Amplifiers , 4x32-bit timers, 24x16-bit PWM counters, 2xQEI, 1xECAP, supports Real-Time Clock (RTC) , Built-in Die Temperature Sensor (DTS)	Dual-Core 480MHz ARM Cortex-M7 & 240MHz Cortex-M4 MCU, 2 Mbytes embedded Flash memory, 2 ESC Integrated PHY, USB HS OTG, Additional RMII/MII Ethernet MAC with IEEE 1588 for multiprotocol support, TFT-LCD display controller, Security and Cryptographic Accelerator , 96-bit UID, Watchdog, RTC /SysTick timers, Rich communication/control interfaces such as SPI/UART/I2C /I2S/SAI/CAN/SDMMC/ADC/DAC/HDMI-CEC/PWM/DFSDM, etc.	BGA routable with standard PCB	Various license models and OpenCore Plus are available. A wide range of Altera FPGAs are supported	Various license models and evaluation Version are available. A wide range of Xilinx FPGAs are supported	estimated market release 4th quarter 2025 Various license models and evaluation Version are available. A wide range of Microchip FPGAs are supported
Further information	https://www.artinchip.com/p/32.html	https://www.artinchip.com/p/32.html				https://www.beckhoff.com/ET1100			
Data Sheet	https://www.artinchip.com/p/32.html	https://www.artinchip.com/p/32.html							

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EtherCAT SubDevice Controller (ESC) Overview

as of July 2025

Name	ET1825/ET1826/ET1827	CF1103	CF1106	GD32H75E	GD32H77E	GDSCN832	Anybus NP40	netX 500	netX 51
Type	Lattice FPGA + IP Core	A4h	A4h	MCU	MCU	ASIC	ARM MPU	ASIC	ASIC
Supplier	BECKHOFF	码灵半导体 Codefair Semiconductor	码灵半导体 Codefair Semiconductor	GigaDevice	GigaDevice	GigaDevice	HMS	hilscher empowering communication	hilscher empowering communication
Supplier	Beckhoff Automation	Codefair Semiconductor Technology	Codefair Semiconductor Technology	GigaDevice	GigaDevice	GigaDevice	HMS Industrial Networks	Hilscher Gesellschaft für Systemautomation mbH	Hilscher Gesellschaft für Systemautomation mbH
Package	FPGA dependent	QFN88L/QFN100L	QFN64L/QFN88L	BGA240	BGA176, LQFP176, BGA100	QFN64	BGA VF400 0.8 mm pitch	BGA345 1 mm pitch	PBGA324 1 mm pitch
Size	FPGA dependent	10mm x 10mm/12mm x 12mm	9mm X 9mm/10mm x 10mm	14 x 14 mm	10 x 10 mm, 26 x 26 mm, 8 x 8mm	9 x 9 mm	17 x 17 mm	22 x 22 mm	19 x 19 mm
µC Interface	serial/parallel (8- /16-/32-bit, async) AMBA-AXI4-Interface	-	/8/16 bit µC interface	OSPI	OSPI/parallel (8/16-bit)	OSPI/parallel (8/16-bit)	Anybus interface (8- / 16-bit 30 ns parallel, 20 MHz SPI, Shift register, UART)	µC bus (internal, 32bit)	µC bus (internal, 32bit)
Digital I/O	8-64*	-	8bit/32bit	-	-	0-16	256 / 256 (Shift register mode)	-	-
General Purpose I/O	0-128*	16bit	16bit	0-116	0-132	0-19	-	16	32
DPRAM	0...60 kByte*	8KB	8KB	8 KB	8 KB	8 KB	12 kByte	256/512 Byte (Mailbox/Process Data)	6 kByte
SyncManager Entities	0...16*	8	8	8	8	8	4	4	8
FMMU Entities	0...16*	8	8	8	8	8	4	3	8
Distributed Clock Support	yes*	64bit	64bit	yes (64 Bit)	yes (64 Bit)	yes (64 Bit)	yes	yes	yes
No. of Ports	1-4 (MII/RMII/RGMII)	2	3	2 (100BaseTX)	2 (100BaseTX)	2 (100BaseTX) + 1 (MII)	2 (MII)	2 (100BaseTX)	2 (100BaseTX)
Specials	estimated market release 4th quarter 2025 Various license models and evaluation Version are available. A wide range of Lattice FPGAs are supported	Integrated MCU（Cortex-M3）	Integrated 2 100Mbps Ethernet PHYs	EtherCAT® SubDevice Controller integrated with Cortex®-M7 600MHz mircocontroller and dual ENET PHY 1207 DMPIS, 1MB SRAM, 4MB FLASH Math acceleration engine, EXMC I/F for FPGA/CPLD extension, HPDF for sigma-delta demodulation PWM, CAN-FD, USB, 100M ENET, rich analogs (14bit ADC, DAC, CMP) TFT-LCD I/F, Image Processing Accelerator, IEC61508 support, Tj 125C, advanced security	EtherCAT® SubDevice Controller integrated with Cortex®-M7 600MHz mircocontroller and optional dual ENET PHY high performance embedded memory (1.1MB SRAM, 2-10 MB NVM), Math acceleration engine, EXMC I/F for FPGA/CPLD extension, HPDF for sigma-delta demodulation PWM, CAN-FD, USB, 100M ENET, rich analogs (14bit ADC, DAC, CMP) integrated RDC and Encoder I/F (Tamagawa, SSI HIPERFACE, EnDAT, Biss-C, Nikon), IEC61508 support, Tj 125C, advanced security	2/3-port EtherCAT® SubDevice Controller with dual integrated ENET PHY which contain a full duplex 100BASE-TX transceiver and support 100Mbps (100BASE-TX) operation, supports HP Auto-MDIX, allowing the use of direct connect or cross-over LAN cables.	Multi-protocol support, ESC Frame forwarding delay: 114 ns, MDP, possible to implement several device profiles	Multi-protocol support, Integrated PHYs, Integrated µC (ARM9-200MHz)	Multi-protocol support, Integrated PHYs, Integrated µC (ARM9-100MHz)
Further information		http://www.codefairsemi.com/product-82239-207809.html	http://www.codefairsemi.com/product-82239-207809.html				www.anybus.com/technologies/network_processors.shtml	https://www.hilscher.com/netx	https://www.hilscher.com/netx
Data Sheet									

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EtherCAT SubDevice Controller (ESC) Overview

as of July 2025

Name	netX 52	netX 90	netX 100	HPM6E00	HPM5E00	XMC4300	XMC4800	LAN9252	LAN9253
Type	ASIC	ASIC	ASIC	RISC-V MPU	RISC-V MPU	ARM MPU	ARM MPU	ASIC	ASIC
Supplier	hilscher empowering communication	hilscher empowering communication	hilscher empowering communication	HPMicro 先楫半导体	HPMicro 先楫半导体	infineon	infineon	MICROCHIP	MICROCHIP
Supplier	Hilscher Gesellschaft für Systemautomation mbH	Hilscher Gesellschaft für Systemautomation mbH	Hilscher Gesellschaft für Systemautomation mbH	HPMicro Semiconductor	HPMicro Semiconductor	Infineon Technologies	Infineon Technologies	Microchip Technology Incorporated	Microchip Technology Incorporated
Package	PBGA244 1 mm pitch	LFBGA144 0.8 mm pitch	BGA345 1 mm pitch	BGA289 (0.8mm pitch) BGA196 (0.8mm pitch)	BGA196 (0.8mm pitch) BGA121 (0.8mm pitch) eLQFP100 (0.5mm pitch)	100 LQFP (0.5 mm)	100 LQFP (0.5 mm)	64 pin QFN (0.5 mm pitch) 64 pin TQFP-EP (0.5 mm pitch)	64 pin QFN (0.5 mm pitch)
Size	15 x 15 mm	10 x 10 mm	22 x 22 mm	14 mm x 14 mm 12 mm x 12 mm	12 mm x 12 mm 9 mm x 9 mm 14 mm x 14 mm	16 x 16 mm	20 x 20 mm 16 x 16 mm 12 x 12 mm	9 x9 mm 12 x 12 mm	9 x9 mm
µC Interface	µC bus (internal, 32bit)	µC bus (internal, 32bit)	µC bus (internal, 32bit)	200MHz µc bus (internal 64bit)	200MHz µc bus (internal 64bit)	µC bus (internal, AHB)	µC bus (internal, AHB)	Host Bus/SPI/SQI	8/16-bit Host Bus/SPI/SQI
Digital I/O	-	-	-	32	32	-	-	0-16*	0-16*
General Purpose I/O	24	16	16	Up to 206	Up to 148	0 - 46	0 - 123	0-16*	0-16*
DPRAM	6 kByte	6 kByte	256/512 Byte (Mailbox/Process Data)	60 kByte	60 kByte	8 kByte	8 kByte	4 kByte	8 kByte
SyncManager Entities	8	8	4	8	8	8	8	4	8
FMMU Entities	8	8	3	8	8	8	8	3	8
Distributed Clock Support	yes	yes	yes	Yes	Yes	yes (64 Bit)	yes (64 Bit)	yes	yes
No. of Ports	2 (100BaseTX)	2 (100BaseTX)	2 (100BaseTX)	3 (MII/RMII/RGMII)	3 (MII/RMII/RGMII)	2 (MII)	2 (MII)	2 (100BaseTX) + opt. 1 (MII)	2 (100BaseTX) + opt. 1 (MII)
Specials	Multi-protocol support, Integrated PHYs, Integrated µC (ARM9-100MHz)	Multi-protocol support, Integrated PHYs, Integrated µC, OnChip Flash 1,5 Mbytes, OnChip DC-DC Converter, (ARM Cortex M4-100MHz) Additional integrated Application Controller (ARM Cortex M4 - 100 MHz)	Multi-protocol support, Integrated PHYs, Integrated µC (ARM9-200MHz)	600MHz Dual-Core RISC-V CPU 2MB SRAM/Opt 4 MB Flash 4-axis motor controll digital encoder interface (EnDat, BiSS, others, etc) Multi-protocol support w/ 3-port 1Gpbs ENET TSN Switch USB HS OTG/CAN-FD UART/SPI/I2C 4x 16b/2MSPS ADC	480MHz RISC-V CPU 512KB SRAM/1 MB Flash 2-axis motor controll USB HS OTG/CAN-FD UART/SPI/I2C 2x 16b/2MSPS ADC	EtherCAT® node on an ARM® Cortex®-M4 processor with up to 256kB on-chip flash, 128kB on-chip RAM and analog/mixed signal capabilities. Qualified for up to 125°C ambient temperature.	EtherCAT® node on an ARM® Cortex®-M4 processor with up to 2MB on-chip flash, 352kB on-chip RAM and analog/mixed signal capabilities. Qualified for up to 125°C ambient temperature.	Cable Diagnostics, 100FX support, 2 integrated PHYs, integrated 1.2V regulator	Host EEPROM Emulation support Supports for low-cost 25MHz crystal Cable Diagnostics, Wake on LAN, 2 integrated PHYs, Single Supply operation (3.3V) Integrated 1.2V regulator
Further information	https://www.hilscher.com/netx	https://www.hilscher.com/netx	https://www.hilscher.com/netx			www.infineon.com/ethercat	www.infineon.com/ethercat		
Data Sheet						www.infineon.com/ethercat	www.infineon.com/ethercat		

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Name	LAN9254	LAN9255	N32H765IxB7EC	N32H785XxB7EC	N32H765IxB7EC	N32H785XxB7EC	i.MX RT1180	EC-1	RZ/T1
Type	ASIC	ARM MCU	ARM MCU	ARM MCU	ARM MCU	ARM MCU	ARM MCU	ARM MPU	ARM MPU
Supplier	Microchip	Microchip	Nation	Nation	NSING	NSING	NXP	Renesas	Renesas
Supplier	Microchip Technology Incorporated	Microchip Technology Incorporated	Nations Technologies Inc	Nations Technologies Inc	NSING Technologies Pte	NSING Technologies Pte	NXP Semiconductors	Renesas Electronics Corporation	Renesas Electronics Corporation
Package	80 pin TQFP-EP (0.5 mm pitch)	128 pin TQFP (0.4 mm pitch)	UFBGA176+25 pitch 0.65mm	TFBGA240+25 pitch 0.8mm	UFBGA176+25 pitch 0.65mm	TFBGA240+25 pitch 0.8mm	289 pin MAPBGA (0.8 mm p.) 144 pin MAPBGA (0.8 mm p.)	196 pin BGA (0.8 mm)	FBGA320 0.8 mm pitch
Size	12x12 mm	14x14 mm	10×10 mm	14×14 mm	10×10 mm	14×14 mm	14 x 14 mm 10 x10 mm	12 x 12 mm	17 x 17 mm
µC Interface	8/16-bit Host Bus/SPI/SQI	SPI/SQI up to 60MHz	µC bus (internal, AHB)	µC bus (internal, AHB)	µC bus (internal, AHB)	µC bus (internal, AHB)		USB Host/Function, CAN, SCIFA, I2C RSPI, Flash	16/32-bit parallel and various serial (SPI/I2C/UART)
Digital I/O	0-32*	0-32*	-	-	-	-		-	-
General Purpose I/O	0-32*	0-32*	126 GPIOs	168 GPIOs	126 GPIOs	168 GPIOs		115* GPIOs / 8 Input (port multiplexed, partial 5V-tolerant, open drain, input pull-up)	0-209*
DPRAM	8 kByte	8 kByte	8KB	8KB	8KB	8KB	8 kByte	512 KB (ATCM) with ECC 32 KB (BTCM) with ECC	8 kByte
SyncManager Entities	8	8	8	8	8	8	8	8	8
FMMU Entities	8	8	8	8	8	8	8	8	8
Distributed Clock Support	yes	yes	yes (64 Bit)	yes (64 Bit)	yes (64 Bit)	yes (64 Bit)	yes	yes (64 bit)	yes
No. of Ports	2 (100BaseTX) + opt. 1 (MII)	2 (100BaseTX) + opt. 1 (MII)	2 (MII)	2 (MII)	2 (MII)	2 (MII)	2 (MII)	2 (MII)	2 (RMII/MII)
Specials	Host EEPROM Emulation support Supports for low-cost 25MHz crystal Cable Diagnostics, Wake on LAN, 2 integrated PHYs, Single Supply operation (3.3V) Integrated 1.2V regulator	Integrated SAME53J ARM Cortex-M4F MCU 1MB Progammable Flash 256KB Main Memory SRAM Extended Industrial Temperature rated (-40 to +105C) Cable Diagnostics, Wake on LAN, 2 integrated PHYs, Single Supply operation (3.3V) Integrated 1.2V regulator	600MHz ARM Cortex-M7 MCU, 2/4MB on-chip FLASH, 1504KB SRAM + 4KB Backup SRAM, 3 12-bit 5Msps ADCs, 4 high-speed comparators, 6 12-bit DACs, integrated with multiple high-speed U(S)ART, I2C, xSPI, SPI, USBHS Dual Role, CAN-FD, SDRAM, FEMC, SDMMC, Ethernet, supports digital camera interface (DVP), supports TFT-LCD graphical interface, JPEG hardware encoder/decoder, and GPU, built-in high-performance encryption algorithm hardware acceleration engine, supports AES/TDES, SHA, SM4 algorithms, supports TRNG true random number generator, supports CRC8/16/32.	Dual-Core 600MHz ARM Cortex-M7 & 300MHz Cortex-M4 MCU, 2/4MB on-chip FLASH, 1504KB SRAM + 4KB Backup SRAM, 3 ADCs, 4 high-speed comparators, 6 DACs, integrated with multiple high-speed U(S)ART, I2C, xSPI, SPI, USBHS Dual Role, CAN-FD, SDRAM, FEMC, SDMMC, Ethernet, supports digital camera interface (DVP), supports TFT-LCD graphical interface, JPEG hardware encoder/decoder, and GPU, built-in high-performance encryption algorithm hardware acceleration engine, supports AES/TDES, SHA, SM4 algorithms, supports TRNG true random number generator, supports CRC8/16/32.	600MHz ARM Cortex-M7 MCU, 2/4MB on-chip FLASH, 1504KB SRAM + 4KB Backup SRAM, 3 12-bit 5Msps ADCs, 4 high-speed comparators, 6 12-bit DACs, integrated with multiple high-speed U(S)ART, I2C, xSPI, SPI, USBHS Dual Role, CAN-FD, SDRAM, FEMC, SDMMC, Ethernet, supports digital camera interface (DVP), supports TFT-LCD graphical interface, JPEG hardware encoder/decoder, and GPU, built-in high-performance encryption algorithm hardware acceleration engine, supports AES/TDES, SHA, SM4 algorithms, supports TRNG true random number generator, supports CRC8/16/32.	Dual-Core 600MHz ARM Cortex-M7 & 300MHz Cortex-M4 MCU, 2/4MB on-chip FLASH, 1504KB SRAM + 4KB Backup SRAM, 3 ADCs, 4 high-speed comparators, 6 DACs, integrated with multiple high-speed U(S)ART, I2C, xSPI, SPI, USBHS Dual Role, CAN-FD, SDRAM, FEMC, SDMMC, Ethernet, supports digital camera interface (DVP), supports TFT-LCD graphical interface, JPEG hardware encoder/decoder, and GPU, built-in high-performance encryption algorithm hardware acceleration engine, supports AES/TDES, SHA, SM4 algorithms, supports TRNG true random number generator, supports CRC8/16/32.	Multi-protocol support 240 MHz ARM® Cortex®-M33 Core (RT1181) 240 MHz ARM® Cortex®-M33 Core plus 800 MHz ARM® Cortex®-M7 Core (RT1189)	Safety Functions, Multi-Function Pin Controller	Additional Ethernet port (RMII/MII), 2-axis high-speed motion control support, digital encoder interfaces (EnDat, BiSS, others), Multi-protocol support, security option, functional safety support, Cortex-R4F (450/600MHz), Cortex-M3 (150MHz) cores
Further information	https://www.microchip.com/en-us/product/LAN9254	https://www.microchip.com/en-us/product/LAN9255	https://www.nationstech.com/product/general/n32h765xx/	https://www.nationstech.com/product/general/n32h785xx/	https://nsing.com.sg/product/General/cortex/7%32H765xB7EC/	https://nsing.com.sg/product/General/cortex/7%32H785xB7EC/	https://www.nxp.com/products/i.MX-RT1180	www.renesas.com/en-eu/ec-1	www.renesas.eu/products/mpumcu/rz/index.jsp
Data Sheet			https://www.nationstech.com/support/dow/	https://www.nationstech.com/support/dow/	https://nsing.com.sg/api/dowfilebefore?did=3003	https://nsing.com.sg/api/dowfilebefore?did=3005	https://www.nxp.com/docs/en/data-sheet/IMXRT1180EC.pdf		www.renesas.eu/products/mpumcu/rz/Documentation.jsp

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EtherCAT SubDevice Controller (ESC) Overview

as of July 2025

Name	RZ/N2L	RZ/T2M	R-IN32M3-EC	RX72M	C2000™ (TMS320F28388D/S)	Sitara AMIC110 SoC	Sitara AM3357/9	Sitara AM4377/9	Sitara AM571xE
Type	ARM MPU	ARM MPU	ARM MPU	ARM MPU	TI C28x subsystem(s) with ARM Connectivity Manager	ARM MPU	ARM MPU	ARM MPU	ARM MPU
Supplier	RENESAS	RENESAS	RENESAS	RENESAS	TEXAS INSTRUMENTS	TEXAS INSTRUMENTS	TEXAS INSTRUMENTS	TEXAS INSTRUMENTS	TEXAS INSTRUMENTS
Supplier	Renesas Electronics Corporation	Renesas Electronics Corporation	Renesas Electronics Corporation	Renesas Electronics Corporation	Texas Instruments Incorporated	Texas Instruments Incorporated	Texas Instruments Incorporated	Texas Instruments Incorporated	Texas Instruments Incorporated
Package	FBGA225, 0.8mm pitch FBGA121, 0.8mm pitch	FBGA320, 0.8mm pitch FBGA225, 0.8mm pitch	BGA324 1 mm pitch	LFBGA224, 0.8mm pitch LFBGA176, 0.8mm pitch LFQFP176, 0.5mm pitch LFQFP144, 0.5mm pitch LFQFP100, 0.5mm pitch	337 BGA 0.8mm pitch 176 QFP 0.5mm pitch	324-pin NFBGA 0.8mm pitch	324-pin NFBGA 0.8 mm pitch	491-pin NFBGA, 0.65mm pitch (0.8 mm effective routing)	760-pin FCBGA 0.8 mm pitch
Size	13 x 13 mm 10 x 10 mm	17 x 17 mm 13 x 13 mm	19 x 19 mm	LFBGA224: 13 x 13 mm LFBGA176: 13 x 13 mm LFQFP176: 24 x 24 mm LFQFP144: 20 x 20 mm LFQFP100: 14 x 14 mm	16 x16 mm 26 x 26 mm	15x15mm	15 x 15 mm	17 x 17 mm	23 x 23 mm
µC Interface	USB Host/Function, CAN-FD, SPI, SCI, DSMIF, I2C, xSPI, External BUS IF(SRAM/SDRAM/Host IF)	USB Host/Function, CAN-FD, SPI, SCI, DSMIF, I2C, xSPI, External BUS IF(SRAM/SDRAM)	16/32-bit parallel (master/slave) and serial (SPI/I2C/UART)	USB, CAN, UART, SPI, I2C, SCI, QSPI	16-bit async PDI interface	200MHz interconnect (internal, 32bit)	200 MHz interconnect (internal, 32bit)	200 MHz interconnect (internal, 32bit)	200 MHz interconnect (internal, 32bit)
Digital I/O	-	-	-	44	N/A	8	8	8	8
General Purpose I/O	0-134*	0-193*	0-96*	0-182*	32	>32	> 32	> 32	> 32
DPRAM	8 kByte	8 kByte	8 kByte	8 kByte	16 kByte	8 kByte	8 kByte	28 kByte	28 kByte
SyncManager Entities	8	8	8	8	8	8	8	8	8
FMMU Entities	8	8	8	8	8	8	8	8	8
Distributed Clock Support	yes	yes	yes	yes (64 bit)	yes	yes	yes	yes	yes
No. of Ports	3 (RGMII/RMII/MII)	3 (RGMII/RMII/MII)	2 (100BaseTX)	2 (100BaseTX/MII/RMII)	2 (MII)	2 (MII)	2 (MII)	2 (MII)	2 (MII)
Specials	Multi-protocol support (EtherCAT, etc), Optimized for network companion chip (parallel bus slave, xSPI slave interface to connect external application CPU), One chip solution for various applications, Functional safety support, Cortex-R52 (400MHz) core	Multi-protocol support (EtherCAT, etc), Optimized for motor control (2-axis high-speed motion control support, digital encoder interfaces (EnDat, BiSS, others, etc.), Functional safety support, Cortex-R52 Dual (800MHz) cores	Multi-protocol support, SPI, I2C, UART, 1.3 Mbyte int. RAM, <1W typical incl. 2 PHYs	Multi-protocol support (EtherCAT, etc.), Security option, Encryption option, 105 °C operating temperature support, Functional safety support	EtherCAT slave enabled real-time controller. Up to 925 MIPS. Single or dual C28x + CLA control subsystems for real-time control loops. Arm based Connectivity Manager for communictions and host control. On-chip flash, RAM, 4x 16-bit ADC, SDFM, 32-ch PWM, analog comparator subsystem, multiple communications ports, configurable logic block for CPLD/FPGA replacement and absolue encoder support.	Entire EtherCAT slave controller can be implemented on internal memory (no external DDR needed), Industrial Communications Subsystem (PRU-ICSS) for multi-protocol support, CAN	Industrial Communications Subsystem (PRU-ICSS) for multi-protocol support, Gigabit Switch, CAN, display, ARM Cortex-A8 (275MHz-1000MHz)	Multi-protocol support, Second PRU-ICSS for Motor control (EnDat, sigma delta filtering etc), Gigabit Switch, CAN, Display subsystem, 2D/3D graphics, Camera I/F, Optional secure boot, ARM Cortex-A9 (upto 1 GHz)	Dual Industrial Communications Subsystem (PRU-ICSS) for multi-protocol support (2 EtherCAT slave instances or EtherCAT slave to protocol gateway), Motor control (EnDat, sigma delta filtering), 2D/3D Graphics, Display subsystem, Video acceleration, PCIe, SATA, Optional secure boot, ARM Cortex-A15 (upto 1.5GHz), 2x M4 cores, 1x C66x DSP core
Further information	www.renesas.com/rzn2l	www.renesas.com/rzt2m	www.renesas.eu/automation		www.ti.com/product/tms320f28388d	www.ti.com/amic110	www.ti.com/AM335x	www.ti.com/AM437x	www.ti.com/product/AM5718
Data Sheet	https://www.renesas.com/us/en/document/dst/rzn2l-group-datasheet	https://www.renesas.com/us/en/document/dst/rzt2m-group-datasheet	www.renesas.eu/r-in	www.renesas.com/document/data/r0n-group-datasheet-au1?r=1054481	www.ti.com/lit/gpn/tms320f28388d	http://www.ti.com/product/AMIC110/datasheet	www.ti.com/lit/ds/symlink/am3359.pdf	www.ti.com/lit/ds/symlink/am4379.pdf	www.ti.com/lit/ds/symlink/am5718.pdf

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EtherCAT SubDevice Controller (ESC) Overview

as of July 2025

Name	Sitara AM572xE	Sitara AM65x SoC	Sitara AM64x SoC	Sitara AM243x SoC	Sitara AM263x SoC	TR8211	TR8253	ANTAIOS	TRITON
Type	ARM MPU	ARM MPU	ARM MPU	ARM MCU	ARM MCU	ASIC	ASIC	ARM MPU	ARM MPU
Supplier	Texas Instruments	Texas Instruments	Texas Instruments	Texas Instruments	Texas Instruments	创耀科技	创耀科技	YASKAWA	YASKAWA
Supplier	Texas Instruments Incorporated	Texas Instruments Incorporated	Texas Instruments Incorporated	Texas Instruments Incorporated	Texas Instruments Incorporated	Triductor Technology (Suzhou) Inc.	Triductor Technology (Suzhou) Inc.	Yaskawa Europe GmbH	Yaskawa Europe GmbH
Package	760-pin FCBGA 0.8 mm pitch	784-pin S-PBGA 0.8mm pitch	441-pin FCBGA 0.8mm pitch	441-pin FCBGA 0.8mm pitch / 293-pin FCCSP 0.5mm pitch via channel array	324-pin NFBGA 0.8mm pitch	TQFP-100, 14x14mm, 0.5pitch / TFBGA-128, 10x10mm, 0.8pitch	QFN-64, 9x9mm, 0.5pitch / LGA-64, 9x9mm, 0.5pitch	TFBGA-380 (0.65 mm pitch) TFBGA-385 (0.8 mm pitch)	FCBGA-784 (0.8 mm pitch)
Size	23 x 23 mm	23mmx23mm	17.2mmx17.2mm	17.2mmx17.2mm / 11mmx11mm	15mmx15mm	TR8211: 14×14 mm TR8211K: 10×10 mm	TR8253: 9x9 mm TR8253L: 9x9 mm	15 mm x 15 mm 19 mm x 19 mm	23 mm x 23 mm
µC Interface	200 MHz interconnect (internal, 32bit)	250MHz interconnect (internal, 256bit)	250MHz interconnect (internal, 128bit)	250MHz interconnect (internal, 128bit), SPI (external)	200MHz interconnect (internal, 64bit)	SPI/SQI Slave HBI (8/16-bit Multiplexed/Indexed)	SPI/SQI Slave HBI (8/16-bit Multiplexed/Indexed)	SPI / QSPI / 16 Bit asynchronous interface	SPI / QSPI / 16 Bit asynchronous interface
Digital I/O	8	8	8	8	8	Aug 32	0-16	26Bits Input, 20Bits Output	32Bits Input, 22Bits Output
General Purpose I/O	> 32	>32	>32	>32	>32	0-16	0-16	up to 32	up to 32
DPRAM	28 kByte	60 kByte	60 kByte	60 kByte	28 kByte	8 kByte	8 kByte	up to 64 kByte	up to 64 kByte
SyncManager Entities	8	8	8	8	8	8	8	8	8
FMMU Entities	8	8	8	8	8	8	8	8	8
Distributed Clock Support	yes	yes	yes	yes	yes	yes (64-bit)	yes (64-bit)	yes (64 bit)	yes (64 bit)
No. of Ports	2 (MII)	2 (MII)	2 (MII)	2 (MII)	2 (MII)	2-3 (MII)	2 (100BaseTX) + opt. 1 (MII)	2 (100BaseTX) or 2 (MII)	4 Gbit Ethernet port
Specials	Dual Industrial Communications Subsystem (PRU-ICSS) for multi-protocol support (2 EtherCAT slave instances or EtherCAT slave to protocol gateway), 2D/3D Graphics, Display subsystem, Video acceleration, PCIe, SATA, Optional secure boot, 2x ARM Cortex-A15 (upto 1.5 GHz), 2x M4 cores, 2x C66x DSP cores	Entire ESC can be implemented on internal memory (no external DDR needed), 3x Gigabit Industrial Communications Subsystem (PRU_ICSSG) for multi-protocol support (up to 3 EtherCAT slave instances), PRU_ICSSG also supports Motor Control functionality (Encoder feedback such as Hiperface-DSL and EnDat and Sigma Delta filtering), up to 4x Arm Cortex-A53 cores at 1.1GHz, 2x Cortex-R5F core at 400MHz with optional lock-step for functional safety or other purposes, 2MB on-chip SRAM	Entire ESC can be implemented on internal memory (no external DDR needed), Dual Gigabit Industrial Communications Subsystem (PRU_ICSSG) for multi-protocol support (2 EtherCAT slave instances or EtherCAT slave to protocol gateway), PRU_ICSSG also supports Motor Control functionality (Encoder feedback such as Hiperface-DSL and EnDat and Sigma Delta filtering), up to 4x Arm Cortex-R5F cores at 800MHz, up to 2x Cortex-A53 cores at 1GHz, 1x Cortex-M4 core at 400MHz for functional safety or other purposes, 2MB on-chip SRAM	Entire ESC can be implemented on internal memory (no external DDR needed), Dual Gigabit Industrial Communications Subsystem (PRU_ICSSG) for multi-protocol support (2 EtherCAT slave instances or EtherCAT slave to protocol gateway), PRU_ICSSG also supports Motor Control functionality (Encoder feedback such as Hiperface-DSL and EnDat and Sigma Delta filtering), up to 4x Arm Cortex-R5F cores at 800MHz, 1x Cortex-M4 core at 400MHz for functional safety or other purposes, 2MB on-chip SRAM	Entire EtherCAT slave controller can be implemented on internal memory, Industrial Communications Subsystem (PRU_ICSS), PRU_ICSS also supports Motor Control functionality (Encoder feedback such as Hiperface-DSL and EnDat and Sigma Delta filtering), up to 4x Arm Cortex-R5F cores at 400MHz, 2MB on-chip SRAM	Configurable clock output for external MCU, Configurable MII TX signal phase shift (0-30ns), Single Supply operation (3.3V), Integrated voltage regulator for logic core/PLL	Built-in 100Mbps fast Ethernet PHYs (Support for automatic detection and switching of cross lines i.e. HP Auto MDIX), 25MHz clock output for external MCU, LED polarity and EEPROM size configuration supported, Single Supply operation (3.3V)	Multi fieldbus protocol support, 2 x integrated PHYs, 1 x integrated GBit Ethernet MAC, Integrated ARM® Cortex®-A5 (288MHz), Backplane communication: SliceBus master for profichip's SNAP+ ASIC, Integrated technology module (2xSII / 4xPWM / 4xCounter), QuadSPI interface (e.g. NOR Flash for firmware), DDR2 external memory interface, Other external interfaces: SD/MMC, NAND, USB2 device, SRAM master/slave, SPI master/slave	Multi fieldbus protocol support, 2 port Real-Time Ethernet switch with integrated PHYs, 2x integrated GBit Ethernet MAC,3 Integrated ARM® Cortex®-A17 (1.26GHz), Secure Core, Backplane communication: SliceBus master for profichip's SNAP+ ASIC, Integrated technology module (2xSII / 4xPWM / 4xCounter), QuadSPI interface (e.g. NOR Flash for firmware), DDR4 external memory interface, 3 PCI Express® Controller, Other external interfaces: 3 SD/MMC, NAND, USB2 device, SRAM master/slave, SPI master/slave
Further information	www.ti.com/product/AM5728	www.ti.com/product/am6548	www.ti.com/product/am6442	www.ti.com/product/am2434	www.ti.com/product/am2634	https://www.triductor.com/product-5.html	https://www.triductor.com/product-5.html		
Data Sheet	www.ti.com/lit/ds/symlink/am5728.pdf	http://www.ti.com/product/am6548/datasheet	http://www.ti.com/product/am6442/datasheet	http://www.ti.com/product/am2434/datasheet	http://www.ti.com/product/am2634/datasheet	https://www.triductor.com/product-5.html	https://www.triductor.com/product-5.html		

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